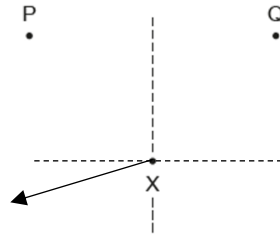


- 9 (a) similarity: both are radial or both have inverse square variation with distance **B1**
- difference: direction is always/only towards the mass or
direction can be towards or away from charge **B1**
- (b) gravitational force = $mg = 1.67 \times 10^{-27} (9.81) = 1.64 \times 10^{-26} \text{ N}$ **A1**
- electric force = $qE = \frac{qV}{d} = \frac{1.6 \times 10^{-19} (270)}{1.8 \times 10^{-2}}$ **C1**
- $= 2.4 \times 10^{-15} \text{ N}$ **A1**
- electric force is about 10^{11} times larger than gravitational force **B1**
- hence gravitational field not taken into consideration **A0**
- (c) (i) $V = \frac{Q}{4\pi\epsilon_0 r}$ **C1**
- $\frac{4.0 \times 10^{-9}}{4\pi\epsilon_0 x} + \frac{-7.2 \times 10^{-9}}{4\pi\epsilon_0 (0.12 - x)} = 0$
- $4.0 (0.12 - x) = 7.2x$
- $x = 0.043 \text{ m}$ **A1**
- (ii) At P, the electric field strengths due to both charges are in the same direction **B1**
- so the electric field strength is not zero. **A1**

- (iii) straight arrow drawn leftwards from X, in direction between extended line joining Q and X and the horizontal **A1**



- (d) The gravitational field strength equals the negative of the gravitational potential gradient. **B1**

- (e) (i) gravitational potential is zero at infinity **B1**
 gravitational force is attractive **B1**
 mass getting closer to moon/planet loses potential energy or negative work done to bring mass from infinity to moon/planet **B1**

- (ii) Any two points below, 1 mark each **B2**
- potential at surface of planet is smaller than at surface of moon
 - potential gradient at surface of planet is smaller than at surface of moon
 - magnitude of potential varies inversely with distance from centre near the spheres
 - point of maximum potential is nearer to the moon than planet

- (iii) curve, starting with gradient of decreasing magnitude at $2R$ and finishing with gradient of increasing magnitude at $D - R$ **B1**
 field strength shown as zero at the point of maximum potential **B1**
 negative field strength near one sphere and positive field strength near the other **B1**

